



Virtual design of flexoelectric sensor for electronic skin

Reporting

Project Information

FlexSen

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Biodiversity	☐		

Coordinated by

INTERUNIVERSITAIR MICRO-ELECTRONICA CENTRUM
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Periodic Reporting for period 1 - FlexSen (Virtual design of flexoelectric sensor for electronic skin)

Reporting period: 2023-04-01 to 2025-03-31

Summary of the context and overall objectives of the project ▼

The Flexsen project addresses the critical need for advanced sensing technologies by developing a novel flexoelectric-based pressure sensor. Traditional sensors typically rely on piezoelectric materials, which only generate electric polarization in response to mechanical stress under specific conditions. In contrast, flexoelectricity is a universal phenomenon present in all dielectric materials, enabling electric polarization through strain gradients regardless of material symmetry. This unique property offers the potential to create highly sensitive and versatile sensors applicable across a wide range of industries, including healthcare, environmental monitoring, and consumer electronics.

The primary objective of Flexsen is to design, model, and experimentally realize an unconventional electromechanical sensor that leverages the flexoelectric effect. To achieve this, the project employs a multidisciplinary approach, integrating computational modeling with precise micro-fabrication and experimental characterization. The focus on inorganic materials, specifically silicon nitride, marks a significant advancement, as flexoelectricity in non-piezoelectric dielectric materials has not been extensively explored until now.

A cornerstone of the Flexsen project is the development of a comprehensive computational framework that not only models flexoelectricity but also encompasses related phenomena such as piezoelectricity and ferroelectricity. Utilizing advanced finite element methods and innovative machine learning techniques, the project simulates complex electromechanical interactions, facilitating the optimization and accurate prediction of sensor performance. Experimentally, the project successfully fabricated silicon nitride thin films and integrated them into sensor prototypes using state-of-the-art micro-fabrication tools available at the host institute, imec.

One of the project's most significant achievements is the first-time observation of flexoelectric coupling in silicon nitride, an amorphous material previously not known to exhibit this effect. This groundbreaking discovery has profound implications, potentially leading to the development of next-generation nano- and micro-electromechanical systems (NEMS/MEMS) that are more efficient, reliable, and adaptable than their traditional counterparts.

The impact of Flexsen extends beyond scientific advancements. Economically, the development of more sensitive and versatile sensors can drive innovation across various sectors, enhancing product performance and enabling new applications. Societally, improved sensor technology can lead to better healthcare monitoring devices, environmental sensors for pollution control, and more responsive consumer electronics, thereby enhancing quality of life and public safety.

Industrially, the project's outcomes provide valuable tools and methodologies for manufacturers of MEMS and NEMS devices, offering new ways to design and optimize sensors for diverse applications. The integration of machine learning with traditional modeling techniques also sets a precedent for future research, promoting the adoption of artificial intelligence in materials science and engineering.

Furthermore, Flexsen fosters strong collaborations between computational scientists and experimentalists, exemplifying the interdisciplinary efforts necessary to drive forward technological breakthroughs. The project's innovative approach and significant findings position the PI and imec to pursue further research and secure additional funding, ensuring the continued advancement and application of flexoelectric-based technologies.

In conclusion, the Flexsen project not only advances our understanding of flexoelectric phenomena in inorganic materials but also paves the way for practical applications that can transform various industries. The project's success highlights the potential of flexoelectric-based sensors to meet the

evolving demands of modern technology, underscoring the value of sustained investment in fundamental and applied research.

Work performed from the beginning of the project to the end of the period covered by the report and main results achieved so far

1. Computational Modeling of Flexoelectricity:

A robust computational framework was developed to investigate the flexoelectric effect in dielectric materials. This model incorporated strain gradient-dependent energy terms into the governing equations, enabling precise simulation of flexoelectric phenomena. Finite element methods (FEM) with advanced basis functions, such as NURBS and Hermite polynomials, were utilized for accurate numerical analysis. Additionally, a simplified multi-layered flexoelectric beam model was derived, demonstrating practical applicability for both sensing and actuating scenarios. The computational framework extended to modeling related phenomena like piezoelectricity and ferroelectricity and incorporated machine learning techniques for forward and inverse modeling of electromechanical devices.

2. Experimental Design and Parametric Studies

Building on the computational models, parametric studies were conducted to optimize material geometries and boundary conditions. For example, rectangular block geometries were selected over more complex shapes to ensure compatibility with standard micro-fabrication techniques. The computationally efficient multi-layered beam model was instrumental in guiding these studies, enabling rapid evaluation of the flexoelectric coefficient under various conditions.

3. Fabrication of Flexoelectric Samples

Utilizing the state-of-the-art cleanroom facilities at the host institute, imec, the project successfully fabricated flexoelectric sensors using silicon nitride thin films. This involved precise deposition of dielectric and electrode layers via sputtering and plasma-enhanced chemical vapor deposition (PECVD), followed by photolithography and reactive ion etching to pattern the devices. A meticulous fabrication process ensured high-quality samples with well-defined dimensions and electrode configurations, essential for accurate characterization.

4. Electromechanical Characterization

A custom-built three-point bending test setup was designed and implemented to characterize the flexoelectric effect in the fabricated samples. This setup included a piezoelectric actuator, a clamping device with pogo pins for electrical connections, and a transimpedance amplifier for measuring the induced voltage. The system was fully automated using Python, allowing precise control over experimental parameters and efficient data collection. The characterization confirmed the flexoelectric coupling in silicon nitride, marking a significant scientific milestone.

Main Achievements

1. Breakthrough in Flexoelectricity Research

Observation of flexoelectric coupling in silicon nitride, an amorphous material not previously known to exhibit this effect, represents a scientific breakthrough. This discovery opens new avenues for exploring flexoelectricity in other non-piezoelectric materials.

2. Comprehensive computational framework

Development of a versatile FEM-based computational model for electromechanical coupling

phenomena, including flexoelectricity, piezoelectricity, and ferroelectricity. The incorporation of machine learning further enhances the model's predictive and optimization capabilities.

3. Fabrication and Characterization success

Fabrication of high-quality flexoelectric devices using a streamlined micro-fabrication process. The custom-built bending test system enabled accurate measurement of flexoelectric parameters, validating both the theoretical models and experimental design.

4. Impact on Future Research Applications

The methodologies and findings from the Flexsen project provide a foundation for the development of advanced MEMS/NEMS devices. The success of this research lays the groundwork for future investigations into flexoelectricity and its applications in sensors, actuators, and beyond.

Progress beyond the state of the art and expected potential impact (including the socio-economic impact and the wider societal implications of the project so far)

The Flexsen project has produced several groundbreaking results that advance the state of the art in electromechanical sensing technologies. These outcomes encompass computational innovations, experimental achievements, and scientific discoveries, with potential implications across multiple disciplines and industries.

1. Scientific Discovery: Flexoelectricity in Silicon Nitride

For the first time, the flexoelectric effect has been observed in silicon nitride, an amorphous, non-piezoelectric material widely used in microelectronics and photonics. This discovery challenges existing assumptions about flexoelectric phenomena, previously limited to crystalline materials, and opens a new frontier for research and applications in flexoelectricity.

2. Advanced Computational Framework for Electromechanical Coupling

The project developed a robust finite element modeling (FEM) framework capable of analyzing flexoelectricity, piezoelectricity, and ferroelectricity. This framework incorporates higher-order differentiations and strain gradient-dependent terms to model complex electromechanical interactions. The inclusion of machine learning techniques, such as physics-informed neural networks, enhances the capability for both forward and inverse modeling, making the tool adaptable for a range of MEMS/NEMS devices.

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